THE COLLATZ COVERING CLAIM AS AN EXPLICIT HYPOTHESIS: A MACHINE-CHECKED LEAN 4 SPECIFICATION

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Abstract. We give a Lean 4 specification of the Collatz map $T(n) = n/2$ for even n and $T(n) = 3n+1$ for odd n, of its iterates, and of the *covering claim* that every positive integer reaches 1 under iteration. The single certified theorem states that the covering claim implies that every positive integer lies in the reverse tree of 1. The claim enters as an explicit hypothesis of the theorem; in an earlier version of the repository it was a user-declared axiom. The result is conditional and makes no unconditional statement about the Collatz conjecture. The development builds with no `sorry`, no `native_decide` and no user axioms, and `#print axioms` reports no axioms at all.

1. Introduction

The Collatz conjecture asserts that for every positive integer n some iterate of T equals 1. It is open. The repository `collatz-covering-claim` records the conjecture in Lean 4 as a named proposition, `RemainingLemma`, so that downstream developments can state results conditionally on it. An earlier version introduced `RemainingLemma` through the declaration `axiom remaining_lemma_assumption : RemainingLemma`. That made every downstream theorem depend on an unproved axiom of the Lean environment. In the present version the axiom is removed and the claim is an explicit hypothesis. A theorem that uses it shows the hypothesis in its statement, and `#print axioms` reports nothing beyond Lean's standard axioms.

2. Definitions

Definition 2.1 (`CoveringClaim.T`, `CoveringClaim.iter`)**.** For $n \in \mathbb{N}$ let $T(n) = n/2$ if n is even and $T(n) = 3n + 1$ otherwise. Let $T^0(n) = n$ and $T^{k+1}(n) = T(T^k(n))$.

Definition 2.2 (`CoveringClaim.InReverseTree`, `CoveringClaim.RemainingLemma`)**.** A natural number n is *in the reverse tree* of 1 if $T^k(n) = 1$ for some $k \in \mathbb{N}$. The *covering claim* `RemainingLemma` is the proposition that every $n > 0$ is in the reverse tree of 1.

3. Result

Theorem 3.1 (Conditional; `CoveringClaim.covering_from_assumption`)**.** *Assume the covering claim* `RemainingLemma`*. Then for every $n \in \mathbb{N}$ with $0 < n$ there exists k with $T^k(n) = 1$.*

Proof. The conclusion is, by definition, the statement of `RemainingLemma`. The Lean proof returns the hypothesis. □

Remark 3.2. Theorem 3.1 is a tautology: hypothesis and conclusion coincide after unfolding definitions. Its value is that the Collatz conjecture is stated precisely and checked by Lean, so later work can cite it as a hypothesis and Lean tracks that dependency. No partial progress towards the conjecture is claimed.

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4. Correspondence between claims and Lean declarations

Every mathematical claim of this paper is listed below with the Lean declaration that proves it. "Conditional" means the hypothesis named in the statement is an explicit argument of the Lean theorem and is *not* proved in the repository. No claim in this paper goes beyond these statements.

Claim	Lean declaration	Status
Covering claim $\Rightarrow$ every $n > 0$ reaches 1	`CoveringClaim.covering_from_assumption`	Conditional on `RemainingLemma`

5. Scope and limitations

The Collatz conjecture (`RemainingLemma`) is not proved here, and nothing in the repository bears on its truth. The single theorem is conditional on it. Earlier descriptions of this repository mention a 2-adic cylinder covering and a file `CoveringClaim/Defs.lean`. That file does not exist, and no statement about residue classes or 2-adic cylinders is formalized. Statement change in this version: `remaining_lemma_assumption` is no longer an axiom. It is now the explicit hypothesis of `covering_from_assumption`.

Appendix A. Lean source

The complete Lean source of the certified development (copyright headers omitted) is reproduced verbatim.

`CoveringClaim/Claim.lean`.

```
/-
Claim.lean
-/

/-
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The covering equation. Stated, not proved.
-/

namespace CoveringClaim

def T (n : Nat) : Nat :=
  if n % 2 = 0 then n / 2 else 3 * n + 1

def iter : Nat → Nat → Nat
  | 0, n => n
  | k + 1, n => T (iter k n)

def InReverseTree (n : Nat) : Prop :=
  ∃ k : Nat, iter k n = 1

def RemainingLemma : Prop :=
  ∀ n : Nat, 0 < n → InReverseTree n

/-- Conditional form: the former axiom `remaining_lemma_assumption` is now an
```

```
explicit hypothesis. This theorem proves nothing about Collatz unconditionally. -/
theorem covering_from_assumption (remaining_lemma_assumption : RemainingLemma) :
    ∀ n : Nat, 0 < n → InReverseTree n :=
  remaining_lemma_assumption

end CoveringClaim
```

Appendix B. Verification certificate

The formal development was machine-checked as follows.

- Repository:
  `https://github.com/BenFrohman/collatz-covering-claim`
- Verified commit:
  `1c6c87f35e2f11af4e669b8c84ac41def9fd568e`
- Lean toolchain: `leanprover/lean4:v4.34.0`
- Mathlib: none (the development depends only on Lean core)
- Continuous integration run (build, axiom audit, `sorry`/`axiom` scan; passed):
  `https://github.com/BenFrohman/collatz-covering-claim/actions/runs/41794903507`
- SHA-256 of the GitHub source tarball of the verified commit, downloaded twice with identical result:
  `https://github.com/BenFrohman/collatz-covering-claim/archive/1c6c87f35e2f11af4e669b8c84ac41def9fd568e.tar.gz`
  `83b67b66ff30c9c984e76a98f0d8c4cbfa156e4f8bf19645bb3f8c1556375287`

The tarball hash refers to the verified commit rather than to a release tag, since a document cannot contain the hash of an archive that includes itself. The Lean sources are unchanged between that commit and the release tag.

The development contains no `sorry` and no user-declared axioms. Every one of the 1 theorems and lemmas listed below depends only on Lean's standard axioms `propext`, `Classical.choice` and `Quot.sound` (or on none). The table is the verbatim output of `lake env lean AxiomAudit.lean` (`#print axioms`) on the verified commit.

Lean declaration	`#print axioms`
`CoveringClaim.covering_from_assumption`	`(no axioms)`

References

[1] J. C. Lagarias, *The $3x+1$ problem and its generalizations*, Amer. Math. Monthly 92 (1985), 3–23.
[2] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, CADE-28, LNCS 12699 (2021), 625–635.

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